

ME381 ROBOTICS

Experiment No.: 6

Experiment Title: Kinematics of a two-link serial planar manipulator

Group No.: B6

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Results for the group task:

2. Forward Kinematics

Notation:

θ_1 : base motor angle, θ_2 : elbow motor angle, l_1 : proximal (base) link, l_2 : distal(second) link

The robot link lengths are: $l_1=200$, $l_2=200$ (mm)

The equations of the forward kinematics of 2R:

$$X = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$Y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

Sr. no.	Base motor (degrees)	Elbow motor (degrees)	Calculated coordinates (X, Y) (units: mm)	Measured coordinates (X, Y) (units: mm)
1	0	0	(400, 0)	(398, 15)
2	90	-90	(200, 200)	(204, 210)
3	-90	90	(200, -200)	(212, -193)
4	-80	30	(163.28, -350.17)	(201, 330)
5	60	-30	(273.21, 273.21)	(277, 283)
6	20	-10	(384.90, 103.13)	(380, 116)
7	-30	30	(373.21, -100)	(376, -84)
8	0	45	(341.42, 141.42)	(337, 169)

The possible reasons for the difference in the measured coordinates and the calculated coordinates are:

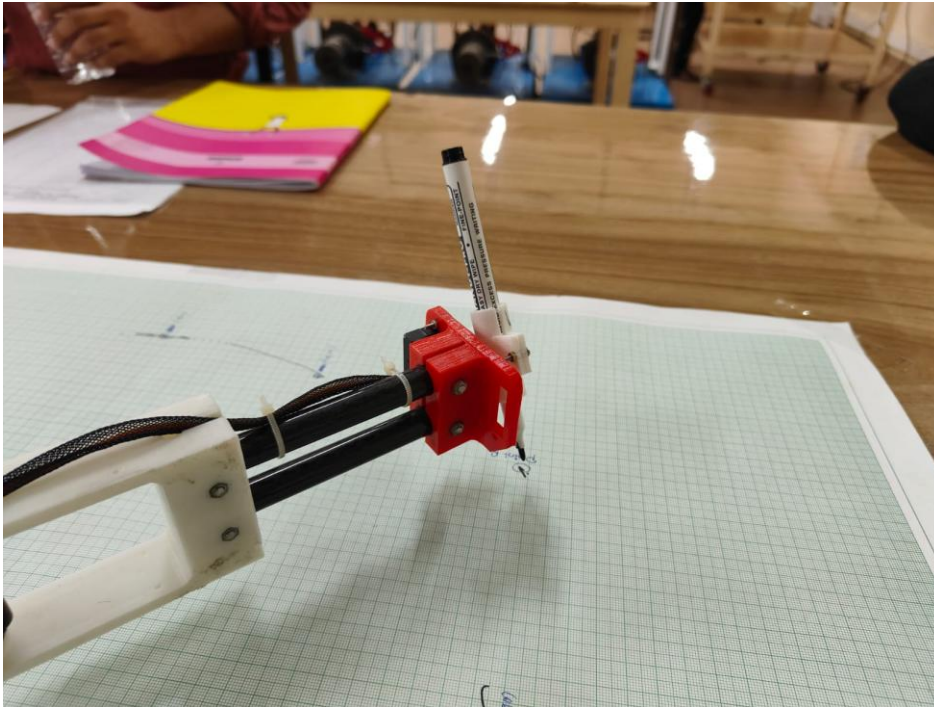
- Robot base or graph paper may not have been perfectly aligned to robot's coordinate frame.
- Graph paper or robot base may move slightly or the table may not be perfectly flat.
- Encoders measure angles in steps, so readings are rounded and cause small errors in position.

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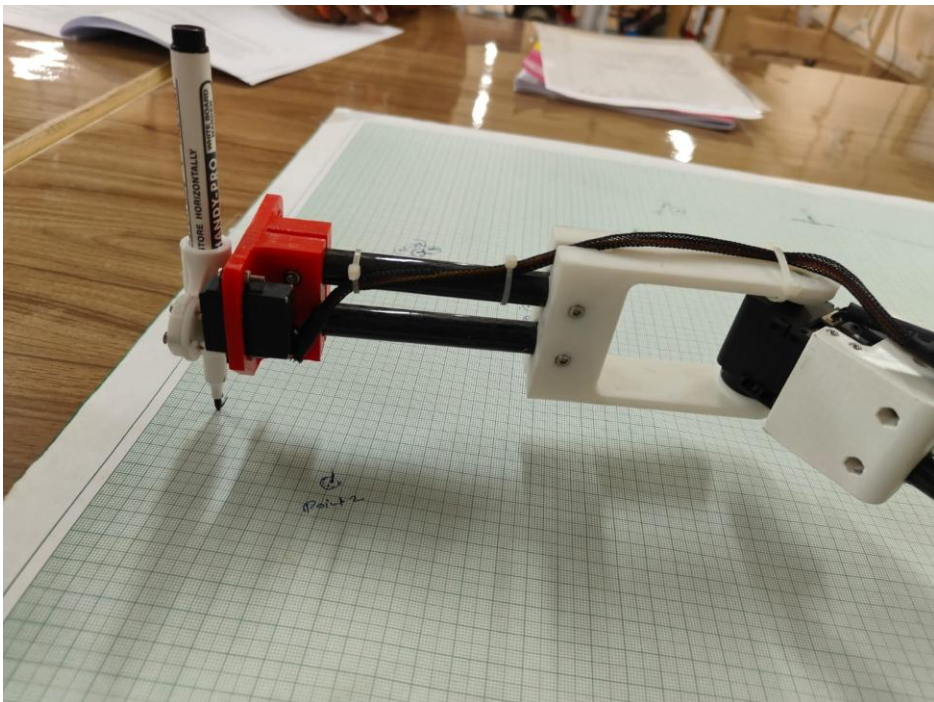
3. Teach and Toggle

1. Two photos of the taught (recorded) positions:

First Point:



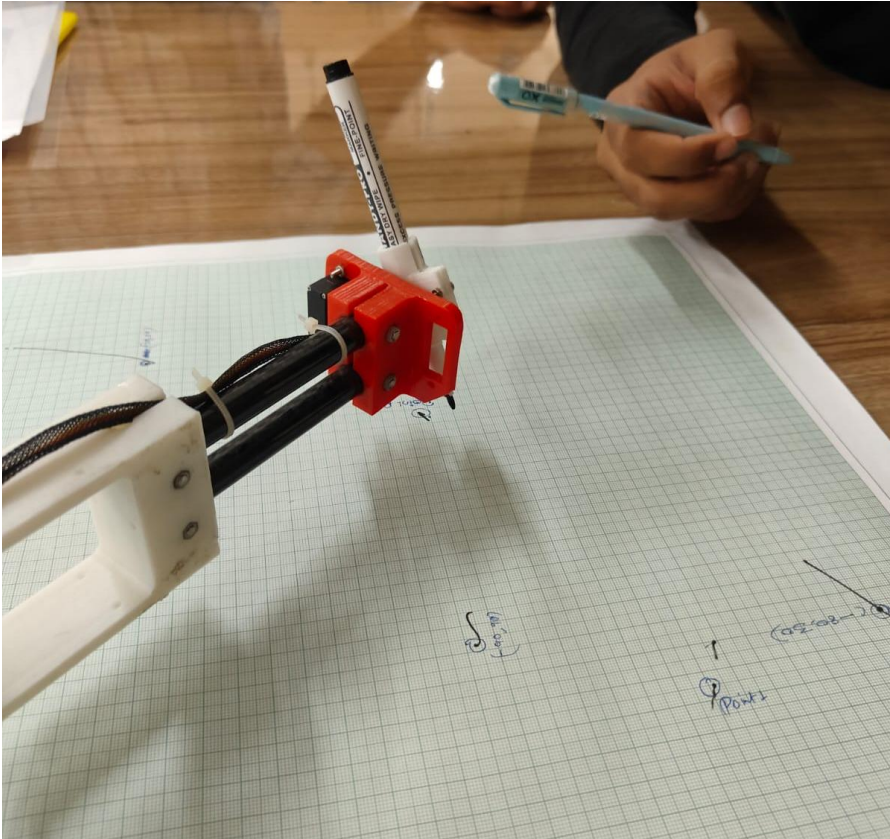
Second Point:



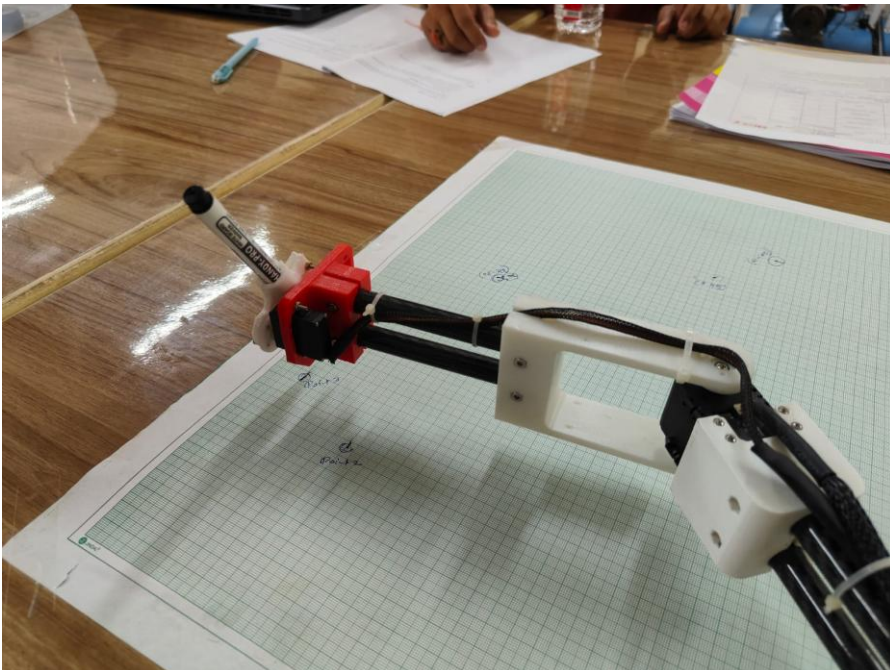
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2. Two photos when the robot reaches those positions (achieved positions)

First Point:



Second Point:



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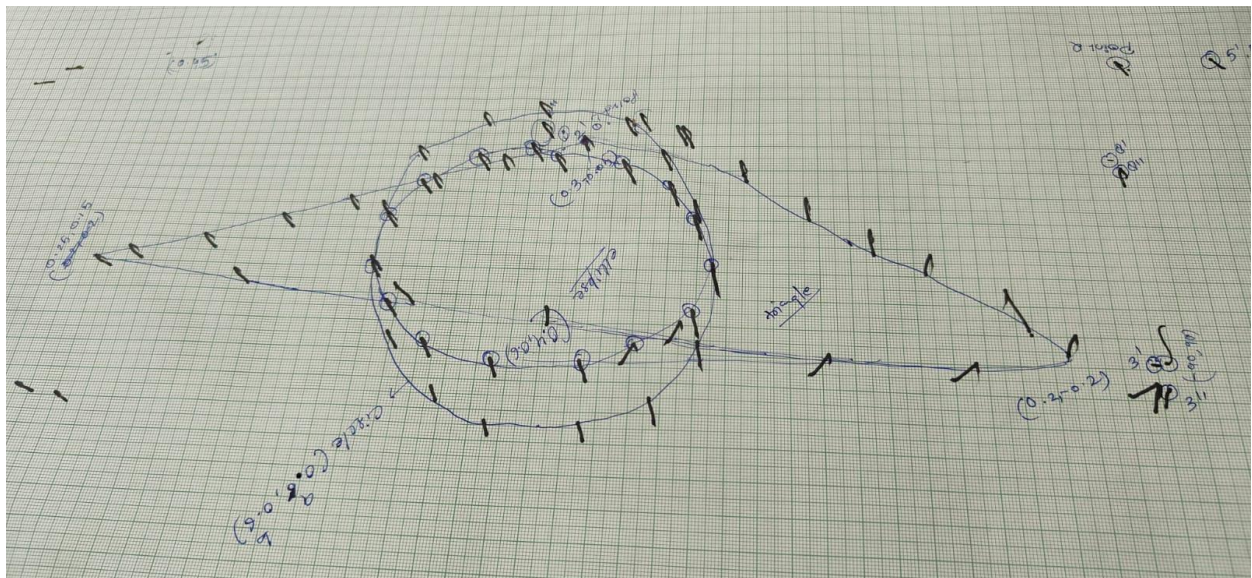
4. Inverse kinematics

Sr. No.	X(m)	Y(m)	Solution 1: (θ_1, θ_2) in degrees	Solution 1: (ϕ_1, ϕ_2) in degrees	Remarks (if any)
1	0.2	0.2	(0.507, 89.736)	(89.756, -87.097)	Point inside workspace
2	0.3	0.0	(-40.672, 82.405)	(41.935, -80.532)	Point inside workspace
3	0.2	-0.2	(-88.503, 89.443)	(0.587, -88.27)	Point inside workspace
4	0.3	-0.3	(-44.282, -0.293)	(-44.282, -0.293)	Point outside workspace
5	0.4	-0.3	(-36.07, 0.0)	(-36.07, 0.0)	Point outside workspace
6	0	0.4	(90.029, 0)	(90.029, 0)	Point on the boundary
7	0.25	-0.25	(-72.141, 55.132)	(-16.716, -53.372)	Point inside workspace
8	0.277	-0.2	(-65.982, 62.17)	(-4.106, -61.29)	Point inside workspace

Reasons for getting one solution is that either the point lies on the boundary of the workspace, both solutions merge into one, or outside the workspace.

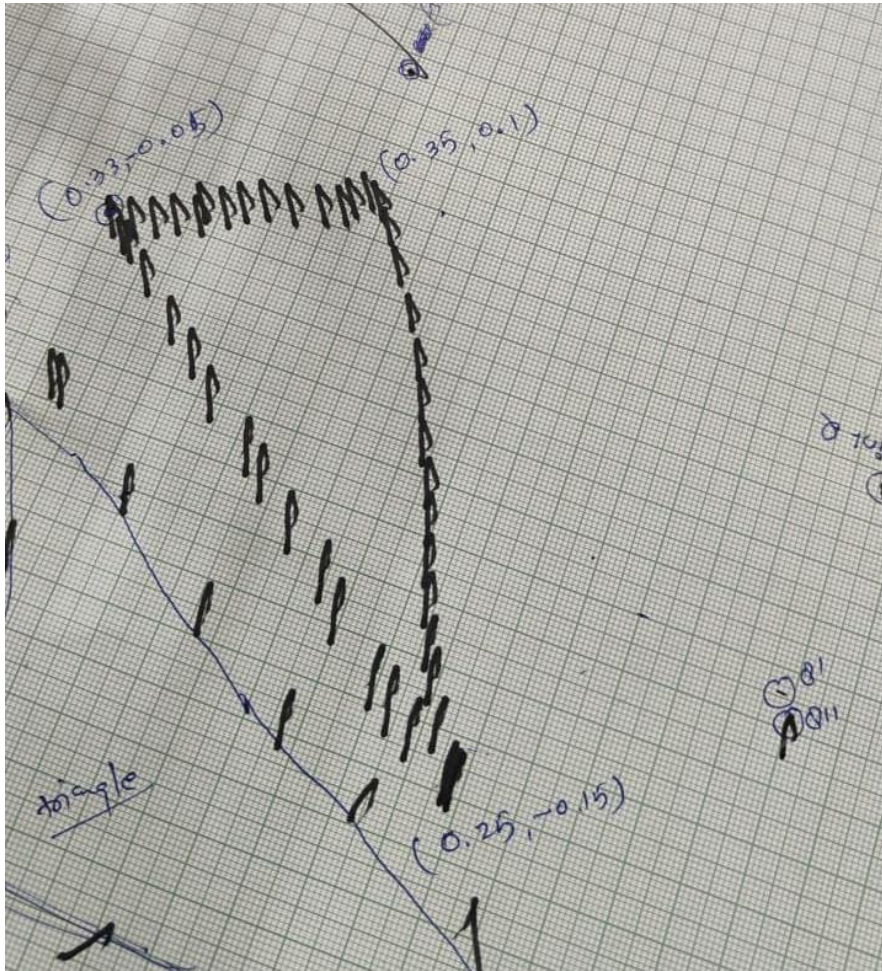
5. Inverse kinematics: trajectory tracking

Image from code1:



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Image from code2:



I. In code 1, ellipse is being tracked, and in code 2, triangle is being tracked.

II.

For code 1:

Ellipse Equation

$$x(\alpha) = 0.25 + 0.04 \cos(\alpha)$$

$$y(\alpha) = 0.25 + 0.04 \cos(\alpha)$$

Specifics of the curve

- Shape: Ellipse
- Center: (0.25 m, 0 m)
- Semi-major axis (x-direction): 0.04 m (40 mm)

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- Semi-minor axis (y-direction): 0.06 m (60 mm)
- Parametrization: α from $0 \rightarrow 2\pi$
- Discretization: 10 points around the ellipse

For code 2:

The manipulator is tracing a **triangle** with vertices at:

- $P1 = (0.25, -0.15)$
- $P2 = (0.33, -0.05)$
- $P3 = (0.35, -0.10)$

Specifics of the curve

- Shape: Closed polygonal curve (triangle).
- Vertices: $(0.25, -0.15)$, $(0.33, -0.05)$, $(0.35, -0.10)$.
- Number of edges: 3.
- Points per edge: 15